# Resume

**Chenyin Chu**

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## Education

**Doctor/Master of Philosophy (Mar 2023 – Current),** specialized in Neuroscience, Alzheimer's disease, The University of Melbourne, Monash University

**Master of Science (Mar 2021- Dec 2022),** specializing in Mathematic physics and Applied mathematics, The University of Melbourne

**Bachelor of Science (Sep 2016 - Sep 2020),** specialized in applied mathematics, Harbin institute of technology, China

## Research Leadership Award

**Melbourne Research Scholarship:** The Melbourne Research Scholarship is highly competitive, with only 300 recipients selected globally. In this round, applicants must have a weighted average mark (WAM) exceeding 92 to be considered. (2024)

**Alzheimer's Association International Conference Fellowship**, 2023, 2024

**National Research Leadership Student Award**, China Ministry of Industry and Information Technology, 2018

## Employment

Jun 2017 - Sep 2019, 0.6 FTE, supervised by Prof. Wenxue Li Contribution: project design, perform data analysis, manuscript writing

## Publications

**1. C. Chu**, Y. Wang, L. Ma, et al. Development and validation of a predictive model to forecast the cerebrovascular risk burden in older people using the NACC and ROSMAP dataset. ***Alzheimer's & Dementia***. 2025; 21: e70543. https://doi.org/10.1002/alz.70543

**2. C. Chu**, Y. Wang, et al., Development and validation of a model to predict the progression of Alzheimer's disease, ***Age and Ageing***, Volume 54, Issue 7, July 2025, afaf198, https://doi.org/10.1093/ageing/afaf198

**3. C. Chu**, Y. Wang, L. Ma, et al. Developing and validating a prediction tool for cerebral amyloid angiopathy neuropathological severity. ***Alzheimer's & Dementia***. 2025;21: e14583. https://doi.org/10.1002/alz.14583

**4. C. Chu**, Y. Wang, L. Ma, et al. Predicting Severity of Cerebral Amyloid Angiopathy Neuropathology: A

Modeling Approach using NACC and ROSMAP Data. ***Alzheimer's & Dementia***. 2025; 21:e70057. https://doi.org/10.1002/alz.70057

**5. C. Chu**, Y. Wang, Y. Wang, et al. Development and validation of a tool to predict onset of mild cognitive impairment and Alzheimer dementia. ***JAMA Network Open***. 2025;8(1): e2453756. doi:10.1001/jamanetworkopen.2024.53756.

**6.** Y. Pan, **C. Chu***, Y. Wang, Y. Wang, Y. Wang, et al. Development and validation of the Florey Dementia Risk Score web-based tool to screen for Alzheimer's disease in primary care. 2024. ***eClinicMedicine***, 76, doi: 10.1016/j.eclinm.2024.102834. *Co-first author, corresponding author

**7. C. Chu**, Y. Low, L. Ma, et al. How can we use mathematical modeling of amyloid-β in Alzheimer's disease research and clinical practices? 2023. Journal of Alzheimer's Disease, 97(1):89-100 doi: 10.3233/JAD-230938.

**8.** L. Ma, Y. Low, Y. Zhuo, **C. Chu**. et al. Exploring the association between cancer and cognitive impairment in the Australian Imaging Biomarkers and Lifestyle (AIBL) study. 2024. ***Scientific Reports***, 14:4364, doi: https://doi.org/10.1038/s41598-024-54875-3.

**9.** Y. Wang, S. Liu, A.G. Spiteri, A. Huynh, **C. Chu**. et al. Understanding machine learning applications in dementia research and clinical practice: a review for biomedical scientists and clinicians. 2024. ***Alzheimer's Research Therapy*** 16:175 doi: https://doi.org/10.1186/s13195- 024-01540-6.

**10. C. Chu**, L. Ma, Y. Zhuo, et al. Developing a classifier to screen for mild cognitive impairment and Alzheimer's disease using machine learning approaches. 2023. AAIC conference 19: e074760, doi: https://doi.org/10.1002/alz.074760.

**11.** L. Ma, **C. Chu**, Y. Zhuo, et al. Unpuzzling the relationship between hypertension, stroke, and Alzheimer's disease–an observation from the Australian Imaging Biomarkers and Lifestyle Study. 2023. AAIC conference 19: e074743, doi: https://doi.org/10.1002/alz.074743.

**12.** L. Ma, Y. Zhuo, Y. Lin, **C. Chu**, C.L. Master, L. Jin, Y. Pan. Exploring the link between mood disorders and Alzheimer's disease in the Australian Imaging, Biomarker & Lifestyle Study of Ageing (AIBL). 2023. AAIC conference, 19: e082853, doi: https://doi.org/10.1002/alz.082853.

**13.** Y. Xu, **C. Chu**, W. Li, Stabilisation of coupled delayed regime-switching diffusion with continuous-state-dependent switching via intermittent control. 2019. IET Control Theory & Applications, 13(12), doi: https://doi.org/10.1049/iet-cta.2018.6335.

**14.** Y. Xu, **C. Chu**, W. Li, Quantized feedback control scheme on coupled systems with time delay and

distributed delay: A finite-time inner synchronization. 2018. Applied Mathematics and Computation, 337, doi: https://doi.org/10.1016/j.amc.2018.05.022.

## Skills

**MATLAB:** Proficient in MATLAB, with expertise in numerically solving complex ODEs and PDEs. Skilled in building mathematical models across various domains, including mathematical biology, neural networks, and traffic networks. Experienced in analyzing the dynamic behavior of ODE systems, particularly in coupled networks.

**R:** Proficient in R, with strong skills in data cleaning, data handling, and statistical model development. Experienced in developing R Shiny applications for interactive data visualization and analysis.

**Python:** Proficient in Python, with expertise in machine learning applications in medical research, including data preprocessing, model development, and evaluation.

**Overleaf:** Proficient in Overleaf for professional document preparation and collaboration. Skilled in LaTeX for creating research manuscripts, theses, and presentations with high-quality formatting. Experienced in managing large projects with multiple figures, tables, and references, ensuring reproducible and publication-ready outputs.